Aviation applications

Four parallel problem types from the Spring 2026 Quiz 11 format. Confirm this semester's coverage in Canvas.

Name: _____ Class: _____

Show a labeled sketch, the equation, the time conversion, and the final answer with units. Use degree mode. Keep full precision until final rounding.

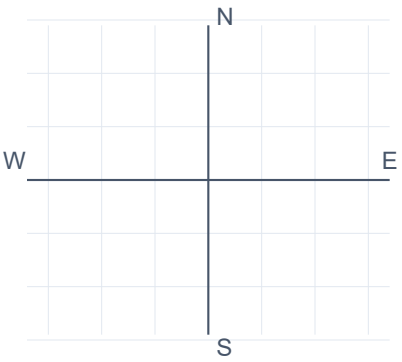
1. An airplane climbs at 9° at an average airspeed of 360 mph along its flight path. How long does it take to gain 5 miles of altitude? Round to the nearest minute.

2. An airplane descends at an average airspeed of 260 mph along its flight path, at an angle of depression of 3°. How much altitude does it lose in 8 minutes? Round to the nearest tenth of a mile.

Bearings and the return flight

For each problem, sketch the route on the compass axes before choosing sine, cosine, or inverse tangent.

3. An airplane travels at 280 knots on a bearing of S38°W (218° compass heading) for 1.8 hours. How many nautical miles south and west has it traveled? Round each to the nearest nautical mile.

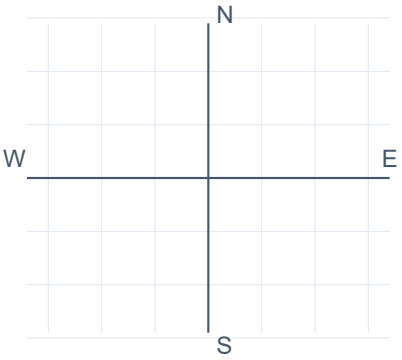


Total distance = _____

South = _____ nautical miles

West = _____ nautical miles

4. An airplane leaves an airport, flies north for 1.5 hours, then west for 0.75 hour, at 350 mph on both legs. Find the direct distance and bearing back to the airport. Round miles and the bearing angle to one decimal.



North leg = _____; west leg = _____

Direct distance back = _____ miles

Return bearing = _____

Check: the return route points toward the airport, opposite both components of the outbound displacement.

Worked answers

Read the setup as well as the final number. Bearings are measured from north or south unless another reference direction is stated.

1. Time to altitude

Vertical speed = 360 sin 9° miles/hour. Time = 5/(360 sin 9°) hours. Multiply by 60: 5.327 minutes, so **5 minutes**. The 360 mph speed is along the sloped flight path, not vertically upward.

2. Altitude lost

Time = 8/60 hour. Flight-path distance = 260(8/60) = 34.666… miles. Altitude lost = 260(8/60) sin 3° = 1.8143 miles, so **1.8 miles**. Use sine to take the vertical component.

3. Both compass components

Total distance = 280(1.8) = 504 nautical miles. Bearing S38°W is measured 38° from south toward west. South = 504 cos 38° = 397.157, so **397 nautical miles south**. West = 504 sin 38° = 310.293, so **310 nautical miles west**.

4. Distance and bearing back

North leg = 350(1.5) = 525 miles; west leg = 350(0.75) = 262.5 miles. Direct distance = $\sqrt{(525^2 + 262.5^2)}$ = **587.0 miles**. The return displacement points south and east. Angle from south = arctan(262.5/525) = 26.5651°. Return bearing: **S 26.6° E**. Equivalent forms are E 63.4° S or compass heading 153.4°.